

Robust Cross-Domain Speech-Based Alzheimer’s Disease Detection via Iterative Adversarial Self-Training

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Abstract

As Alzheimer’s disease (AD) has increasingly become a major global public health issue, speech-based AD detection has attracted widespread attention. However, most existing methods are trained and evaluated on a single dataset, often leading to severe cross-domain performance degradation due to reliance on dataset-specific artifacts rather than disease-related speech cues. In real-world applications, reliable Alzheimer’s disease detection requires models that are robust to variations in recording environments, speakers and data collection conditions. To address this challenge, this paper adopts unsupervised domain adaptation to learn robust, domain-invariant feature representations in the absence of target-domain diagnosis labels. On this basis, a novel unsupervised domain adaptation method, Iterative Adversarial Self-Training (IAST), is proposed. Results demonstrate that IAST significantly improves the generalization ability and robustness under various cross-domain settings.¹

1 Introduction

Alzheimer’s disease has become a major global public health issue. According to estimates by Alzheimer’s Disease International, there are currently approximately 50 million people with dementia worldwide, and this number is still growing rapidly, and is expected to triple by 2050 [Alzheimer Europe, 2019]. Alzheimer’s disease is the leading cause of dementia and is rapidly developing into one of the most expensive, deadliest, and most socially burdensome diseases of this century [Alzheimer’s Association, 2023]. Against this background, how to achieve automated detection of Alzheimer’s disease in a highly reliable, low-cost, and easily scalable manner has become an important direction of current research.

In recent years, deep learning has become the dominant approach for speech-based AD detection. Prior work has explored a wide range of architectures and representations, including end-to-end automatic speech recognition frameworks [Lin et al., 2024], transformer-based models for speech-level AD recognition [Dong et al., 2024], and hybrid approaches that incorporate disfluency-related paralinguistic features [Jin et al., 2023]. At the same time, over the past five years, Interspeech

¹We release our codes and models with this submission: https://sleepwalker554.github.io/IAST_website/

and ICASSP have also organized several Alzheimer’s Dementia Recognition through Spontaneous Speech recognition challenges, including the ADReSS [Luz et al., 2020], ADReSSo [Luz et al., 2021], ADReSS-M [Luz et al., 2023], and PROCESS [Tao et al., 2025], attracting participation from many research teams. Collectively, these studies demonstrate that speech-based AD detection models can achieve strong performance when trained and evaluated under matched, single-dataset conditions.

However, in real-world applications, AD detection often needs to cope with different recording environments, device conditions, and data collection processes. Although existing models can achieve high detection performance when trained and evaluated on a single dataset, their performance often degrades substantially when applied to data collected under different conditions, revealing limited robustness. This degradation arises in part because models trained on a single dataset may learn to rely on dataset-specific recording characteristics rather than disease-related speech features. Prior work has provided clear evidence of this behavior: Liu et al. [Liu et al., 2024] demonstrated that models trained using only silent segments from the Pitt Corpus [Becker et al., 1994] can still achieve extremely high detection accuracy on the same dataset. This suggests the presence of a Clever Hans effect, in which models exploit recording-environment artifacts instead of clinically meaningful speech cues.

Since the training conditions and real-world testing conditions often do not match, when the model is applied to different datasets or different recording environments, this reliance on the specific training environment can lead to a significant degradation in recognition performance. Our cross-domain experiments further confirm these limitations: across multiple model architectures, we observe consistent drops in recognition accuracy under cross-domain evaluation, with a maximum decrease of 34.38% (see Table 2). These results indicate that AD detection models trained and evaluated solely on a single dataset exhibit limited cross-domain generalization, significantly constraining their practical applicability. To address this challenge, this paper investigates unsupervised domain adaptation for speech-based AD detection.

Recent advances in self-supervised speech representation learning [Devlin et al., 2019] have shown that large pre-trained models can capture robust and transferable speech features across speakers and recording conditions. In particular, XLSR [Conneau et al., 2021], pre-trained on large-scale multilingual unlabeled speech data, has demonstrated strong performance in AD detection and related speech-based cognitive impairment tasks [Chen et al., 2023, Pérez-Toro et al., 2022]. These properties make self-supervised speech representations especially promising for cross-domain Alzheimer’s disease detection, where recording conditions and data distributions often vary substantially. Motivated by these insights, this work investigates how different levels of speech representation and unsupervised domain adaptation strategies can be combined to improve robustness and generalization in cross-domain settings. Our main contributions are summarized as follows:

- We design three AD detection models with increasing representational capacity, spanning handcrafted acoustic features, pre-trained self-supervised speech embeddings, and task-specific fine-tuning.
- We conduct a systematic cross-domain evaluation of AD detection across multiple models and unsupervised domain adaptation strategies.
- We propose a novel unsupervised domain adaptation method, Iterative Adversarial Self-Training (IAST), which significantly improves cross-domain generalization and robustness.

2 Related Work

2.1 Existing AD Detection Models

Early AD detection studies primarily explored traditional machine learning approaches. For example, the ADReSS [Luz et al., 2020], ADReSSo [Luz et al., 2021] and ADReSS-M [Luz et al., 2023] Challenges adopted support vector machines as a baseline, while Hason et al. [Hason and Krishnan, 2022] used random forests for speech-based AD detection. More recently, deep learning has become the dominant paradigm in AD detection. For example, Lin et al. [Lin et al., 2024] adopted a Seq2Seq-based end-to-end automatic speech recognition framework for Alzheimer’s disease detection. Dong et al. [Dong et al., 2024] proposed a hierarchical attention-free transformer model for speech-level AD recognition. In addition, Jin et al. [Jin et al., 2023] combined disfluency-based paralinguistic features with a complementary ensemble strategy for speech-based Alzheimer’s disease detection, achieving

favorable experimental results. Recent work has also begun exploring large language models and large audio-language models for AD detection. For example, Heitz et al. [Heitz et al., 2025] used GPT-4 to extract semantic features from spontaneous speech transcripts, while Park et al. [Park et al., 2025] applied Chain-of-Thought reasoning and supervised fine-tuning for AD detection.

Although recent studies have explored various architectures for AD detection, the main focus of this paper is not to design a new AD classifier, but to investigate cross-domain robustness and unsupervised domain adaptation. Therefore, we consider three representative model settings for experiments. They cover feature representation methods with different representational capabilities commonly used in AD detection tasks, including handcrafted acoustic features, frozen self-supervised speech representations, and task-adaptive speech representations. By comparing models with different representational capabilities, this paper can more comprehensively evaluate the impact of cross-domain distribution shifts on AD detection models, as well as the broad effectiveness of the proposed domain adaptation method. The specific architectures of these models will be described in detail in Section 3.

2.2 Datasets

Three public English speech datasets relevant to Alzheimer’s disease detection are considered in this study: the Pitt Corpus [Becker et al., 1994], the ADReSS dataset [Luz et al., 2020], and the Lu Corpus. All three datasets are available through DementiaBank [Lanzi et al., 2023] and contain recordings from the Cookie Theft picture description task [Goodglass and Kaplan, 1983], which elicits spontaneous speech and has been widely adopted for Alzheimer’s disease assessment because speech disfluencies and other paralinguistic characteristics provide important indicators of cognitive impairment [Haider et al., 2019].

The Pitt Corpus is one of the most widely used benchmarks for speech-based Alzheimer’s disease detection and contains recordings from both individuals with Alzheimer’s disease (AD) and healthy controls (HC). The ADReSS dataset was released as part of the ADReSS Challenge at Interspeech 2020 and provides age- and gender-balanced Cookie Theft recordings. The Lu Corpus is an English spontaneous speech dataset collected under different recording conditions, making it particularly suitable for cross-domain evaluation. Table 1 summarizes the numbers of AD and HC recordings in each dataset. Detailed source-target configurations and data splits are presented in Section 4.

Table 1: Sample numbers of datasets.

Dataset	Total Num	AD Num	Control Num
Pitt Corpus	552	309	243
ADReSS	156	78	78
Lu	74	38	36

3 Methodology

This section presents the designed AD detection models, traditional unsupervised domain adaptation methods, and the proposed Iterative Adversarial Self-Training (IAST).

3.1 Models

We implement three speech-based Alzheimer’s disease detection models with increasing representational capacity: an eGeMAPS-based model using handcrafted acoustic features, a pre-trained embedding-based model with frozen parameters, and a fine-tuned pre-trained model for the target task. The architectures are detailed below and illustrated in Figure 1. For the specific implementation of the models, please refer to our released code.

eGeMAPS-based model. As shown in Fig. 1(a), each input utterance is first divided into 10 equal-length segments. Then, the OpenSMILE toolkit [Eyben et al., 2010] is used to extract 25-dimensional eGeMAPS [Eyben et al., 2015] acoustic features from each segment, thereby representing each sample as a temporal feature tensor of shape (10, 25). Subsequently, two linear layers are applied to remap the feature dimension. Batch Normalization, ReLU activation, and Dropout are used between

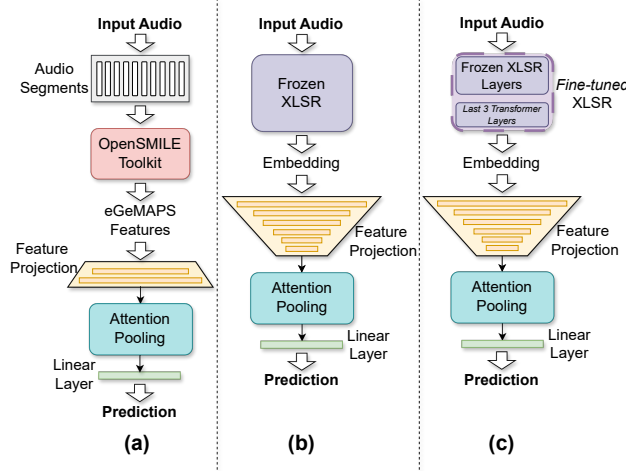


Figure 1: Model architectures: (a) eGeMAPS-based model, (b) frozen-parameter XLSR-based model, and (c) fine-tuned XLSR-based model.

the layers to stabilize training and mitigate overfitting. Next, an attention pooling layer is used to perform weighted aggregation over the 10 segments along the temporal dimension. Finally, a linear classification layer outputs the AD prediction result.

Frozen-parameter XLSR model. As shown in Fig. 1(b), the audio input is first truncated or padded to a fixed length of 60 seconds and then fed into the pretrained XLS-R model [Conneau et al., 2021] with frozen parameters to extract frame-level embedding features. Subsequently, the embeddings are passed through five linear layers, each of which reduces the feature dimension, followed by Batch Normalization and a ReLU activation function. The mapped features are then fed into an attention pooling layer for temporal aggregation. Meanwhile, we introduce a mask mechanism [Vaswani et al., 2017] to mark the padded time steps, ensuring that they do not contribute to the pooled representation. Finally, a linear classification layer outputs the AD prediction result.

Fine-tuned XLSR model. As illustrated in Fig. 1(c), we fine-tune XLS-R using the Pitt Corpus and the ADReSS dataset, respectively. During fine-tuning, to mitigate catastrophic forgetting [French, 1999] and reduce the risk of overfitting, only the last three Transformer layers are updated, while the remaining layers are kept frozen. The classification head is exactly the same as that of the frozen-parameter XLSR model, namely, a structure consisting of five linear dimensionality-reduction layers, mask-based attention pooling, and a linear classification layer.

3.2 Domain Adversarial Training

Domain Adversarial Training (DAT) [Ganin et al., 2016] is used to promote domain-invariant feature learning between the labeled source domain and the unlabeled target domain. During training, the domain discriminator is jointly optimized with the Alzheimer’s disease classification task through adversarial learning. The Alzheimer’s disease predictor is trained to minimize the classification loss on the source domain while maximizing the confusion of the domain discriminator, thereby reducing domain-specific information in the learned representations. This adversarial objective promotes alignment between the feature distributions of the source and target domains without requiring target-domain labels. This training strategy is consistently applied to all three AD detection models described in Section 3.1.

Although DAT can effectively reduce the distributional distance between the source and target domains in the feature space, the model cannot access any labels from the target domain during training. This means that, when updating its parameters, the Alzheimer’s disease classifier learns a decision boundary that is entirely driven by the source-domain distribution.

3.3 Self Training

Self-Training (ST) [Scudder, 1965] is used to leverage unlabeled target-domain data through a pseudo-labeling mechanism. This training strategy is also consistently applied to all three models. The self-training procedure consists of an initialization stage and several learning stages that alternate between pseudo-label generation and retraining. In the initialization stage, the model is first trained using labeled source-domain data and is then used to generate predictions for target-domain samples. In the k -th round, the model f_{k-1} performs Softmax prediction on target-domain samples. If the maximum class probability (confidence) is no lower than τ , then $\arg \max_c \hat{p}_c$ is adopted as the pseudo-label for that sample and combined with the source-domain data for subsequent training. In the experiments, we set $\tau = 0.9$. This process is repeated iteratively, allowing the model to gradually incorporate more reliable target-domain supervisory information, thereby continuously improving its performance on the target domain and enhancing cross-domain generalization.

The effectiveness of self-training depends largely on the discriminative ability of the initial model and the quality of the pseudo-labels. If the initial model has not yet adapted to the target-domain distribution, it will produce many incorrect pseudo-labels on the target domain. As these incorrect labels are repeatedly incorporated into training, the errors continuously accumulate over iterations and eventually deviate from the true target-domain decision boundary.

3.4 Iterative Adversarial Self-Training

Iterative Adversarial Self-Training (IAST) is the unsupervised domain adaptation method proposed in this paper, which combines the advantages of DAT and ST in cross-domain learning while overcoming the two major limitations of DAT lacking target-domain class supervision and ST relying on high-quality initialization. The key idea of IAST is to progressively improve target-domain pseudo-label quality through repeated feature alignment and target-domain adaptation. The proposed pipeline of Iterative Adversarial Self-Training is illustrated in Figure 2. It begins with an initialization stage in which Domain Adversarial Training is applied to align feature distributions between the labeled source domain and the unlabeled target domain. This initial alignment reduces domain-specific biases and yields a more stable model that produces more reliable predictions on target-domain samples. Based on these predictions, Self-Training is then employed to generate pseudo-labels for high-confidence target-domain samples, which are incorporated into subsequent training.

IAST alternates between adversarial feature alignment and self-training in multiple iterations. As domain adversarial training progressively reduces inter-domain feature discrepancies, the quality of the pseudo-labels generated by the model gradually improves. In turn, more reliable pseudo-labels help the model reduce the negative impact of incorrectly classified pseudo-labels during the self-training stage, enabling better adaptation to the target-domain data distribution. Meanwhile, as an increasing number of reliable target-domain samples are incorporated into training, the model’s decision boundary gradually shifts toward the target-domain data distribution, which in turn promotes domain adversarial training to learn more appropriate feature representations. Through this cyclic process, the model is able to simultaneously improve feature alignment and target-domain discriminative capability in each iteration.

The advantage of IAST lies in the complementarity between DAT and ST. DAT first narrows the feature gap between the source and target domains, providing a better initialization for ST and preventing the generation of a large number of incorrect pseudo-labels at the beginning. ST then incorporates category information from the target domain into training by using high-confidence pseudo-labels, addressing the limitation that the task classifier in DAT cannot learn discriminative information from the target domain. Through iterative interaction, the two components reinforce each other, ultimately improving both feature alignment and target-domain classification performance.

4 Experiments

4.1 Experimental Protocol

As described in Section 3, we evaluate speech-based Alzheimer’s disease detection models under one baseline setting and three unsupervised domain adaptation (UDA) settings. We consider two source domains, namely the Pitt Corpus [Becker et al., 1994] and the ADReSS dataset [Luz et al.,

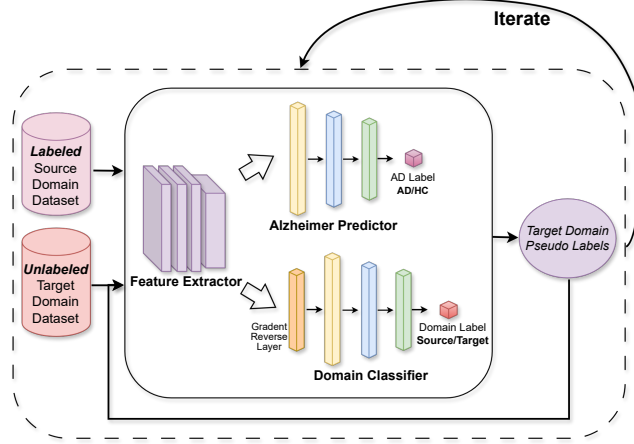


Figure 2: Iterative Adversarial Self-Training pipeline (AD: Alzheimer’s disease; HC: healthy controls).

2020], while the English Lu Corpus serves as the target domain in all experiments. The Pitt Corpus contains 552 speech recordings (309 AD and 243 healthy controls), the ADReSS dataset contains 156 recordings (78 AD and 78 healthy controls), and the Lu Corpus contains 74 recordings (38 AD and 36 healthy controls). All datasets are based on the Cookie Theft picture description task and are obtained from DementiaBank [Lanzi et al., 2023].

For source-domain training, each dataset is divided at the speaker level into 80% training data and 20% validation data. Specifically, the Pitt Corpus provides approximately 442 recordings for source-domain training and 110 recordings for validation, while the ADReSS dataset provides 125 recordings for training and 31 recordings for validation. For the target domain, the Lu Corpus is divided at the speaker level into 80% unlabeled adaptation data (59 recordings) and 20% held-out evaluation data (15 recordings). Speaker overlap is avoided across all splits. In the baseline setting, models are trained using only the labeled source-domain training data and are directly evaluated on the held-out Lu evaluation subset without adaptation. In the UDA settings, the source-domain training data and the unlabeled Lu adaptation subset are jointly used during training. The target-domain adaptation subset is used only for domain alignment or pseudo-label generation, and no target-domain diagnosis labels are accessed during training. Domain Adversarial Training, Self-Training, and the proposed Iterative Adversarial Self-Training are evaluated under the same source-target protocol for fair comparison.

4.2 Model Training Details

In terms of training settings, all models in this paper are trained for AD classification using the cross-entropy loss function and the AdamW optimizer. The learning rate of the *eGeMAPS-based model* is set to 3×10^{-3} . When training on the ADReSS dataset, the batch size is set to 10; when training on the Pitt Corpus, the batch size is set to 16. The learning rate of the *XLSR model with frozen parameters* is set to 3×10^{-3} , and the batch size is set to 32. For the trainable XLSR parameters of the *fine-tuned XLSR model*, a smaller learning rate of 1×10^{-5} is used; for the classification head, a larger learning rate of 1×10^{-3} is used. The batch size is set to 16. For the baseline experiments, we set the maximum number of training epochs to 60 and adopt an early stopping strategy with a patience value of 10.

Based on the above model training settings, we implement three unsupervised domain adaptation methods, including Domain Adversarial Training (DAT), Self Training (ST), and Iterative Adversarial Self-Training (IAST). DAT also adopts a maximum number of training epochs of 60 and an early stopping strategy with a patience value of 10 in each training run. ST first performs one round of initialization training based only on labeled source-domain data, and then performs at most 5 rounds of pseudo-label iterations. IAST first undergoes an initialization training using the DAT method, and then performs at most 5 rounds of iterative training; in each iteration, we combine the target-domain

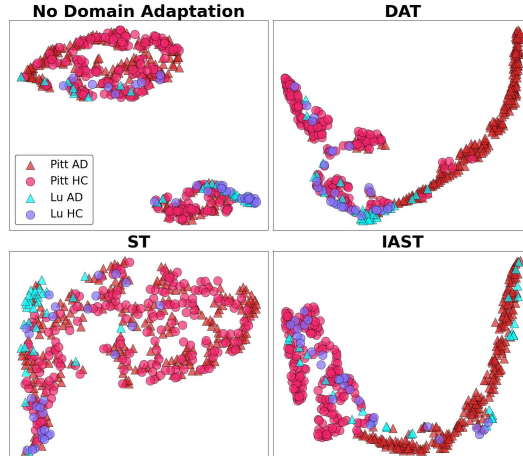


Figure 3: t-SNE visualization of embeddings from the XLSR-based model fine-tuned on Pitt Corpus.

data with pseudo-labels and the labeled source-domain data, and train using the DAT method. For ST and IAST, only target-domain pseudo-labels with confidence greater than or equal to 0.9 are retained in each iteration. A single round of training lasts for at most 60 epochs and adopts an early stopping strategy with a patience value of 10; if fewer than 5 high-confidence pseudo-labels are available in a certain round, subsequent iterations are stopped.

All models were trained on an RTX5090 GPU. We used OpenSMILE (v2.6.0) for eGeMAPS feature extraction, and the software environment included Python 3.9, PyTorch (v2.8.0), and torchaudio (v0.13.1).

4.3 Evaluation Metrics

For evaluation, we report classification accuracy and F1 score, treating Alzheimer’s disease as the positive class. Each experiment is independently repeated using five fixed random seeds 21, 42, 84, 168, 336, and the result with the highest accuracy is reported. In addition, we conduct significance testing on all experimental results.

5 Results and Analysis

Baseline Cross-Domain Performance. Table 2 reports cross-domain results under the baseline setting, where models are trained on a single source dataset and evaluated on the target domain without adaptation. A consistent performance gap is observed across all architectures. For instance, the ADReSS-finetuned XLSR model achieves 84.38% accuracy in-domain but drops to 50.00% on the Lu corpus, which is a 34.38% decrease. Similarly, the Frozen-XLSR model trained on Pitt decreases from 70.27% to 58.11%, and the eGeMAPS-based model drops from 65.62% to 54.05% when transferred from ADReSS to Lu. These results show that strong in-domain performance does not guarantee cross-domain generalization. Even large pretrained models such as XLSR suffer substantial degradation under domain shift, suggesting reliance on source-domain-specific characteristics rather than stable disease-related speech features.

Traditional Domain Adaptation Methods. We next examine the effect of traditional unsupervised domain adaptation methods on cross-domain AD detection performance. As shown in Table 2, both Domain Adversarial Training and Self-Training improve target-domain performance across most model architectures compared to the baseline setting, although the degree of improvement varies substantially by model. For the eGeMAPS-based model trained on the ADReSS dataset, DAT improves target-domain accuracy from 54.05% to 62.50%, while ST yields a similar improvement to 62.50%. When trained on the Pitt Corpus, both DAT and ST again improve target-domain accuracy relative to the baseline, indicating that feature alignment and pseudo-labeling can partially help domain mismatch even for handcrafted acoustic features. More pronounced gains are observed for models based on pre-trained speech representations. For example, the XLSR-based model with frozen

Table 2: Cross-domain performance comparison of speech-based Alzheimer’s disease detection models under different unsupervised domain adaptation methods. (*DAT*: Domain Adversarial Training; *ST*: Self-Training; *IAST*: Iterative Adversarial Self-Training. *Asterisks**: performance significantly different from the binomial majority-class baseline, $p < 0.05$.)

Model	Source Domain	Target Domain	Domain Adaptation	Within Domain		Cross Domain	
				Src Acc	Src F1	Tgt Acc	Tgt F1
eGeMAPS-based model	ADReSS	Lu	–	65.62%	0.6667	54.05%	0.4516
			DAT	62.50%	0.6667	62.50%	0.7000
			ST	68.75%*	0.7222*	62.50%	0.6667
			IAST (proposed)	71.88%*	0.7429*	75.00%*	0.7778*
	Pitt	Lu	–	61.26%*	0.7190*	58.11%	0.6931
			DAT	62.16%*	0.7042*	62.50%	0.7273
			ST	59.46%*	0.7134*	62.50%	0.7273
			IAST (proposed)	58.56%*	0.6462*	68.75%	0.7368
XLSR-based model (Frozen Parameters)	ADReSS	Lu	–	87.50%*	0.8750*	68.92%*	0.6102*
			DAT	81.25%*	0.8421*	81.25%*	0.8235*
			ST	84.38%*	0.8571*	75.00%*	0.7500*
			IAST (proposed)	81.25%*	0.8235*	81.25%*	0.8235*
	Pitt	Lu	–	70.27%*	0.7442*	58.11%	0.6804
			DAT	70.27%*	0.7227*	75.00%*	0.7778*
			ST	71.17%*	0.7576*	81.25%*	0.8421*
			IAST (proposed)	67.57%*	0.7568*	87.50%*	0.8750*
XLSR-based model (Fine-tuned on ADReSS Dataset)	ADReSS	Lu	–	84.38%*	0.8485*	50.00%	0.4789
			DAT	81.25%*	0.8235*	68.75%	0.6154
			ST	84.38%*	0.8387*	81.25%*	0.8235*
			IAST (proposed)	87.50%*	0.8667*	81.25%*	0.8235*
XLSR-based model (Fine-tuned on Pitt Corpus)	Pitt	Lu	–	69.37%*	0.7213*	63.51%*	0.6494*
			DAT	67.57%*	0.7313*	75.00%*	0.7500*
			ST	70.27%*	0.7317*	87.50%*	0.8571*
			IAST (proposed)	70.27%*	0.7660*	87.50%*	0.8571*

parameters trained on the Pitt Corpus improves from 58.11% accuracy on the target domain under the baseline setting to 75.00% with DAT and 81.25% with ST. Similarly, for the XLSR-based model fine-tuned on the ADReSS dataset, ST increases target-domain accuracy from 50.00% to 81.25%. These results suggest that self-training is particularly effective when the underlying model produces sufficiently reliable target-domain predictions, as is often the case for pre-trained representation-based models.

The proposed IAST. We next analyze the effect of the proposed IAST method on cross-domain AD detection. As shown in Table 2, IAST consistently yields the strongest target-domain performance across different model architectures and source–target settings, outperforming or matching the best results achieved by traditional unsupervised domain adaptation methods. For the eGeMAPS-based model trained on the ADReSS dataset, IAST improves target-domain accuracy to 75.00%, substantially exceeding the performance of both DAT and ST (62.50%). When trained on the Pitt Corpus, IAST again achieves the highest target-domain accuracy of 68.75%, compared to 62.50% obtained with DAT and ST. These results indicate that the proposed iterative combination of adversarial feature alignment and self-training is particularly beneficial for models with limited representational capacity. For models based on representations extracted from large-scale pre-trained model, IAST also demonstrates clear advantages. In the case of the XLSR-based model with frozen parameters trained on the Pitt Corpus, IAST increases target-domain accuracy to 87.50%, outperforming both DAT (75.00%) and ST (81.25%). Similarly, for the XLSR-based model fine-tuned

on the ADReSS, IAST improves target-domain accuracy to 81.25%, exceeding the result obtained with DAT and matching the best performance achieved by ST.

Embedding Space. Figure 3 visualizes the embedding distribution of the Pitt-finetuned XLSR model using t-SNE [Maaten and Hinton, 2008]. To quantitatively assess clustering behavior, Table 3 reports two metrics, silhouette score [Rousseeuw, 1987] and separation ratio [Davies and Bouldin, 2009], under different domain adaptation methods. We evaluate two clustering structures: (i) domain separation (Pitt Corpus vs. English Lu Corpus) and (ii) class separation (Alzheimer’s disease vs. healthy controls). The results show that IAST substantially reduces inter-domain discrepancy while simultaneously increasing class separability. This indicates that IAST not only aligns feature distributions across datasets but also enhances discriminative structure in the embedding space.

Table 3: Clustering evaluation from the XLSR-based model fine-tuned on Pitt Corpus (AD: Alzheimer’s disease; HC: healthy controls; SS: Silhouette score; SR: Separation ratio).

Domain adaptation	Pitt vs Lu		AD vs HC	
	SS ↓	SR ↓	SS ↑	SR ↑
–	0.2062	0.9612	0.0189	0.2318
DAT	0.0010	0.9791	0.3707	1.4624
ST	0.1218	0.9747	0.0069	0.1299
IAST (proposed)	0.0031	0.0892	0.4857	1.9484

Source–Target Performance Trade-offs. While unsupervised domain adaptation improves target-domain performance, it can slightly reduce source-domain accuracy. Compared to the baseline, DAT, ST, and IAST occasionally exhibit modest source-domain degradation. For example, for the frozen XLSR model trained on the Pitt Corpus, source-domain accuracy decreases from 70.27% to 67.57% with IAST, while target-domain accuracy increases substantially from 58.11% to 87.50%. This trend reflects a trade-off in which improved cross-domain generalization comes at the cost of reduced specialization to source-domain characteristics, favoring more domain-invariant and robust representations.

6 Conclusion

This study shows that although speech-based Alzheimer’s disease detection models can achieve good performance within a single dataset, they still face a clear performance drop when evaluated across datasets. To improve model robustness in real-world application scenarios, this paper investigates the role of unsupervised domain adaptation methods in cross-domain AD detection and validates them on three types of models: an eGeMAPS-based model, a Frozen-parameter XLSR-based model, and a fine-tuned XLSR-based model.

This paper proposes the Iterative Adversarial Self-Training method. This method continuously and iteratively performs domain adversarial training and self-training to gradually improve the quality of pseudo-labels in the target domain and guide the model to learn more domain-invariant speech features. We conducted experiments on three types of models. The experimental results show that, under different model architecture settings, IAST achieves the best performance on the target domain.

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A Responsible-use Statement

A.1 Limitations

Our research focuses on Alzheimer’s disease speech detection from speech in English contexts. In this field, the vast majority of studies have used the Pitt Corpus [Becker et al., 1994] and its derived datasets, and the number of publicly available English AD speech datasets independent of the Pitt Corpus is very limited. We note that although there are some larger English AD speech datasets, these datasets have not yet been made publicly available and therefore cannot be used for experimental validation in this study.

Other publicly accessible English dementia datasets, such as the VAS Corpus [Liang et al., 2022] and the English WLS Corpus [Herd et al., 2014], are not restricted to AD patients, as they include individuals with MCI or other forms of cognitive decline without distinguishing these diagnostic groups. To ensure the rigor of the research, we did not use these datasets in our experiments. Therefore, due to the limited number of publicly available English AD speech datasets independent of the Pitt Corpus, this paper selects the English Lu Corpus [Lanzi et al., 2023] as the target domain.

A.2 Ethical Considerations

As speech data contain sensitive health-related and personally identifiable information, appropriate privacy and data-governance procedures are necessary. The proposed model should be viewed as a research or screening tool rather than a clinical diagnostic system, and real-world deployment would require broader external validation and professional oversight.

The predictions of the proposed model may remain uncertain when applied to populations, recording environments, or clinical settings that differ from those represented in the datasets used in this study. In particular, the limited number of independent English AD speech datasets constrains our ability

to fully characterize model reliability under broader distribution shifts. Therefore, the reported performance should not be interpreted as evidence of reliable clinical performance across unseen populations or acquisition conditions.

A.3 Licensing

We use the DementiaBank dataset, a shared multimedia database for studying communication in dementia. DementiaBank is part of TalkBank. Therefore, use of the data is generally governed by the Creative Commons Attribution–NonCommercial–ShareAlike 3.0 license (CC BY-NC-SA 3.0), unless otherwise specified. This license requires attribution, restricts commercial use, and requires share-alike for derivatives. Access to DementiaBank is password-protected and restricted to approved members of the DementiaBank consortium. We comply with the DementiaBank data use requirements, including restrictions on redistributing or posting password-protected data on external websites or servers.